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Research paper

Improving student achievement through professional development: Results from a randomised controlled trial of Quality Teaching Rounds

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HIGHLIGHTS

- A four-arm randomised controlled trial examining PD effects on student achievement.
- PD intervention was pedagogy-focused Quality Teaching Rounds (QTR).
- QTR has significant impact on students' mathematics achievement.
- PLCs undertaking peer observation insufficient for growth in achievement.
- QTR is a powerful form of PD with significant potential for wider impact.

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ABSTRACT

Improving student achievement through professional development (PD) is both highly sought-after and elusive. This four-arm randomised controlled trial evaluated effects of Quality Teaching Rounds (QTR), a pedagogy-focused form of PD, on mathematics, reading, and science outcomes for elementary students ($n = 5478$). Outcomes at baseline and 8-month follow-up were compared for QTR, QTR trainer-led, peer-observation, and wait-list control groups. Students in the QTR group made 25% more progress in mathematics than the control group ($g = 0.12$, 95% CI: 0.07–0.17). This result supports QTR as a form of PD with significant potential for wider impact.

1. Introduction

The powerful and lasting influence of teachers on student learning (Chetty et al., 2014; Hattie, 2003; Opper, 2019; Rockoff, 2004; Rowe, 2003; Timperley, 2007) has led to a thriving industry in teacher professional development (PD) worth billions of dollars internationally (Birman et al., 2000; Bowe & Gore, 2017; Garet et al., 2008; Kraft et al., 2018). While improving student outcomes is the fundamental goal of this investment (Kennedy, 2016), the impact of PD is inconsistent and evidence of its effects is often contradictory (Guskey, 2003). Critiques of PD programs centre on the lack of methodological rigour (Guskey & Yoon, 2008), ineffective content, poor design and attention to implementation principles (Hill et al., 2013), and a failure to produce significant changes in teaching practice or student achievement (Darling-Hammond et al., 2017). Evidence of effects on student outcomes is particularly weak (Pellegrini et al., 2018; Scher & O'Reilly, 2009), especially at scale (Hill et al., 2013; Lortie-Forgues & Inglis, 2019). Nonetheless, there is still strong belief in the potential of high quality PD programs to improve student outcomes (Darling-Hammond et al., 2017; Opper, 2019; Timperley, 2007). As a result, the search continues for rigorous evidence of the impact of PD on student achievement (Hill et al., 2013; Kennedy, 2019).

Indeed, since at least the beginning of this century, a substantial body of research has attempted to identify features of PD that lead to significant gains in student achievement. The emergent consensus on essential features of effective PD (Desimone, 2009;

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Penuel et al., 2007) highlights: an emphasis on content (Alton-Lee, 2011; Darling-Hammond et al., 2017; Darling-Hammond & Richardson, 2009; Desimone, 2009; Garet et al., 2001; Scher & O'Reilly, 2009; Timperley, 2007); the provision of active learning opportunities for teachers (Birman et al., 2000; Brownell et al., 2017; Darling-Hammond et al., 2017; Garet et al., 2001), particularly over an extended period of time (Birman et al., 2000; Darling-Hammond et al., 2017; Garet et al., 2001; Kraft et al., 2018); integration of knowledge and skills (Alton-Lee, 2011; Kraft et al., 2018; Timperley, 2007); as well as opportunities for feedback and reflection (Darling-Hammond et al., 2017; Darling-Hammond & Richardson, 2009; Desimone, 2009) and collaboration (Darling-Hammond et al., 2017; Darling-Hammond & Richardson, 2009).

For the most part, however, even programs that incorporate these features of effective PD have been underwhelming in demonstrating effects on student outcomes (Garet et al., 2016; Jacob et al., 2017; Kennedy, 2016; Lortie-Forgues & Inglis, 2019). For example, while a focus on curriculum content has long been regarded as crucial to effective PD (Desimone, 2009; Guskey, 2003; Scher & O'Reilly, 2009; van Veen et al., 2012), improving teachers' content knowledge does not appear to be sufficient for improving student outcomes (Kennedy, 2016). Nonetheless, the privileging of content-focused PD is evident in a recent review of studies, in which 31 of the 35 PD programs examined were centred on enhancing teachers' knowledge of curriculum content (Darling-Hammond et al., 2017).

While only four of the programs emphasised pedagogy, each produced significant effects on student achievement (see Allen et al., 2015, 2011; Antoniou & Kyriakides, 2013; Meyers et al., 2016). Gore & Rosser (2020) contend that the limited volume of research on the effects of pedagogy-focused PD has produced a self-perpetuating cycle; that is, research that has shown limited effectiveness of content-focused PD is often followed by the redoubling of efforts to refine content-focused PD. In this light, we argue that new research investigating the effects of pedagogy-focused PD on student outcomes such as that reported here, is long overdue.

1.1. The professional development approach

Quality Teaching Rounds (QTR) is a pedagogy-focused approach to PD that utilises a form of teaching rounds (Elmore, 2007; Goodwin et al., 2015) undertaken by groups of teachers in professional learning communities (PLCs) (Dufour, 2004). Uniquely, QTR involves teachers observing, analysing, and discussing each other's lessons *guided by* a comprehensive pedagogical model, the Quality Teaching model (QT) (NSW Department of Education and Training (NSWDET), 2006). Derived from work on Authentic Pedagogy (Newmann et al., 1996) and Productive Pedagogy (Gore, 2007; Gore et al., 2004; Lingard et al., 2001; Mills et al., 2009), the QT model treats teaching holistically, honouring the complexity of teaching in ways that resonate with teachers. After two teachers per school attend a two-day (12 h) inservice workshop on QT and QTR, they form PLCs with two (or more) of their colleagues back at school. All school-based rounds are managed by the teachers themselves with no external facilitation or oversight. QTR pays careful attention to power relations among teachers, with reciprocity (every teacher in the PLC hosts a lesson observation and is fully present throughout the process), turn-taking (every teacher explains their views for each element of the QT model), and confidentiality (coding of lesson quality stays within the group) structured into the process (Gore et al., 2017; Gore & Bowe, 2015; Gore & Rickards, 2020; Gore & Rosser, 2020).

While many teachers describe QTR as the most powerful professional development in which they have participated (Gore, 2014), a growing body of statistical evidence, including from a previous RCT, is demonstrating significant effects of QTR on the quality of teaching, teacher morale, and sense of recognition (Gore et al., 2017; Gore, Smith, et al., 2015). These positive effects of QTR hold for both beginning (Gore & Bowe, 2015) and experienced (Gore & Rickards, 2020) teachers, who report greater confidence and stronger professional relationships. The effects apply equally to elementary and secondary teachers (Gore et al., 2017), with enhanced collegiality and professional collaboration also found across typical grade and subject boundaries (Gore & Rosser, 2020). Having obtained these strong results for teachers, this paper shifts attention to the effects of teacher participation on *student outcomes*. We hypothesise that students taught by teachers who participate in QTR will display greater positive achievement in mathematics, reading, and science than students whose teachers do not.

2. Methodology

The study was a cluster randomised controlled trial (RCT) designed to examine the efficacy of teacher participation in QTR for the improvement of student achievement in mathematics, reading, and science. The unit of randomisation was the school. The full methodology, following Consolidated Standards of Reporting Trials (CONSORT) protocols (Moher et al., 2010), is described elsewhere (Miller et al., 2019). In brief, the four-arm parallel RCT involved three intervention groups: Researcher-led QTR (QTR); Trainer-led QTR (QTR-T); and, a Peer Observation time-equivalent active wait-list control group (Peer Observation). A PD-as-usual wait-list control group (Control), whereby schools continued with whatever PD they were already undertaking, formed the reference group against which the interventions were evaluated.

Student achievement in mathematics, reading, and science – the primary outcomes – was measured at baseline and 8-month follow-up. Teachers in the intervention groups undertook their respective two-day workshops and four in-school PD days within a 10-week period after the completion of baseline assessment and random allocation to a study condition. Due to the size of the sample required for this trial ($N = 322$ schools), the study was undertaken as a split cohort design, originally scheduled for the 2019 and 2020 school years. With postponement of the second cohort until 2021 due to the COVID-19 pandemic, this paper reports the findings for students ($n = 5478$) in the first cohort of schools ($n = 125$).

2.1. Participants

Government schools in NSW, Australia, were eligible to participate if they could provide two classes of Stage 2 students (Years 3–4, ages 8–10 years) where the teacher was responsible for the class on a full-time basis, willing to participate, available for the full duration of the study, and had not previously participated in QTR. These teachers were referred to as 'monitored' teachers. All students of eligible and consenting Stage 2 monitored teachers were eligible to participate, except those for whom the school received NSW Department of Education funding for significant identified learning difficulties. While these students were excluded from analysis, they were not excluded from involvement in the class activities associated with the research. A minimum of 15 eligible Stage 2 students was required for a class to participate which was especially pertinent in mixed stage classes. Schools were also required to provide two additional 'PLC teachers' from any stage (K-6) to form a four-person PLC to undertake the randomly allocated intervention. Students of these additional teachers were not invited to participate.

In the case of small schools, there were several scenarios for involvement: 1) a small school could provide one eligible Stage 2 class (teacher and students) and form a PLC with three additional teachers; or 2) two to four schools could combine to form a ‘small school network’ of four teachers, providing there was at least one eligible Stage 2 class. Schools participating in QTR and Peer Observation were compensated for the cost of substitute teachers ($500 per day, per participating teacher), but received no direct financial inducement. Schools in the wait-list control and Peer Observation groups received the same level of funding for participating teachers to undertake QTR in the subsequent year. All PLCs were formed prior to randomisation.

2.2. Interventions

2.2.1. Researcher-led QTR

Quality Teaching Rounds involves sequential sessions that occur on a single day:

1 Reading discussion: Designed to support the group in developing a shared theoretical basis for professional conversations, this session also helps build a sense of professional community (typically 1 h);
2 Observation: One PLC member teaches a lesson that is observed by all other members of the PLC (a full lesson length, typically 30–80 min);
3 Coding: Individual coding of the observed lesson is carried out, including by the observed teacher, for all elements of the Quality Teaching model (NSW Department of Education and Training, 2006); and,
4 Discussion: All PLC members discuss the observed lesson, and teaching more broadly, drawing on the language and concepts of the Quality Teaching model and work towards consensus (typically one to 2 h).

The intent of QTR is to focus teachers’ attention on the relationship between classroom practice and student learning. QTR observations show respect for the teacher and the teaching-learning process by watching a whole lesson each time (Bowe & Gore, 2017). The use of the QT model for observation, analysis, and discussion provides teachers with a common language and set of conceptual standards to foster rigorous diagnostic conversations. The protocols of confidentiality and egalitarian turn-taking seek to flatten power hierarchies and ensure all participants contribute to and are heard in the analysis of the observed lesson and pedagogy in general (Bowe & Gore, 2017; Gore & Bowe, 2015).

The QT model incorporates three dimensions of pedagogy: Intellectual Quality, Quality Learning Environment, and Significance. Each dimension comprises six elements, as shown in Table 1. This pedagogical framework has been widely implemented in Australia and utilised by other researchers who have reported its capacity to transform teachers’ practice (see Aubusson et al., 2007; Ewing et al., 2003; Hammond, 2008; Liberante, 2012; Miller et al., 2017; Moloney & Xu, 2018; Penney et al., 2009; Plummer & Nyholm, 2010; Prieto et al., 2015; Rushton, 2008; Treble, 2009; Whalan, 2012).

Prior to commencing QTR, two of the four teachers from each school in this intervention arm participated in a two-day workshop delivered by the lead researcher from the project team who played a key role in developing both QT and QTR. The workshop provided background information on the QT model and QTR and emphasised the importance of each component of the approach (i.e., readings, PLCs, observations, individual coding, and group discussion). Teachers were provided with opportunities to practise coding using the QT model and to participate in simulated rounds discussions using filmed lesson extracts of 15–20 min’ duration. Those who took part in workshops then formed a PLC back in their schools and led their colleagues in QTR with no further input from the research team other than informal conversations during data collection and implementation fidelity checks.

2.2.2. Trainer-led QTR

In order to examine the potential for implementing QTR PD at scale, the Trainer-led QTR arm enabled comparison of the Researcher-led and Trainer-led delivery of the two-day QTR workshop. Participation in this arm used the same protocols as the Researcher-led QTR arm, and schools received equivalent compensation for teachers’ time away from the classroom.

The trainers were three competitively selected ‘QTR Advisers’ with recent school experience. Using a ‘train-the-trainer’ or ‘cascade’ approach (Hayes, 2000), and the iterative development, testing, and evaluation of processes, they were prepared for delivering the QTR workshops. The Advisers engaged in 25 days of training, led by the developers of QTR, aimed at ensuring they:

- developed a deep understanding of the QT model and its 18 elements;
- gained experience in reliably coding lessons;
- consolidated their knowledge of processes and protocols surrounding QTR;
- observed QTR workshops and QTR in action in schools;
- built their understanding of the power relations involved in QTR delivery; and,
- learned strategies to manage ethical issues and group dynamics.

A developmental evaluation of this training process was conducted prior to the commencement of the RCT, during which the Advisers ran a QTR workshop with teachers from four pilot schools. This evaluation involved workshop observation, in-school observations, and fidelity checks of QTR as well as interviews to evaluate implementation after Adviser-led workshop delivery and to refine the train-the-trainer model.

2.2.3. Active control group (peer observation)

The active control group was included to increase the capacity for causal inference by matching the release time and funding levels provided for the two QTR intervention groups. Research participants randomly allocated to the active control group engaged in a PD alternative to QTR, known as Peer Observation. Several other high-profile evidence-based options were

Table 1
Quality Teaching model (3 dimensions, 18 elements).

Intellectual Quality	Quality Learning Environment	Significance
Deep knowledge	Explicit quality criteria	Background knowledge
Deep understanding	Engagement	Cultural knowledge
Problematic knowledge	High expectations	Knowledge integration
Higher order thinking	Social support	Inclusivity
Metalanguage	Students’ self-regulation	Connectedness
Substantive communication	Student direction	Narrative

considered, including Lesson Study, Instructional Rounds, and Instructional Coaching. Peer Observation (Australian Institute for Teaching and School Leadership [AITSL], 2017) offers several features that made it a suitable alternative to QTR for the active control group. It uses a similar structure (teachers observe teachers in a non-hierarchical manner), requires the same amount of time (four days spread over a period of weeks), and involves the same number of participants in each school (four). As such, it presents the closest form of professional development to QTR, minus the underlying QT pedagogical model and structured observation and discussion processes associated with QTR.

Four teachers per school, forming a PLC, participated in the active control intervention arm. At least two teachers per school participated in a two-day Peer Observation workshop led by a researcher from the project team who had no involvement in the delivery of QTR PD workshops. The workshop provided information on the Peer Observation structure of pre-observation conversation, lesson observation, post-observation conversation, and post-observation action plans (AITSL, 2017). Participants were also provided with a widely used set of tools for note-taking before and during observations from the University of Toronto's Centre for Teaching Support and Innovation (Centre for Teaching Support & Innovation, 2020). Those who took part in the workshops then led the Peer Observation professional development intervention back in their schools with no further input from the research team, other than informal conversations held during data collection and implementation fidelity checks.

2.3. Outcomes

2.3.1. Primary outcome

Student achievement, the primary outcome, was measured using the most recent editions of the Progressive Achievement Tests (PATs) for Mathematics (PAT-M version 4), Reading (PAT-R version 5), and Science (PAT Science) (Australian Council of Educational Research [ACER], 2011). These measures have been rigorously tested and are widely used in Australian schools and internationally to monitor students' skills, understandings, and growth over time. Item reliability produced using Rasch modelling is reported at 0.87–0.91 (Lindsey et al., 2010), with this measure interpreted in the same manner as Cronbach's alpha within classical test theory (Bond & Fox, 2007). Fogarty (2007) reports that these tests have good predictive validity when administered to 805 Australian secondary students. In each of the subject areas we chose developmentally appropriate tests for Year 3 and Year 4 students. Scaled scores allow for comparison of students across test levels and testing years.

Three test levels were available in reading and mathematics, while in science there was only one test per grade. In order to mitigate ceiling and floor effects, we used the middle test at baseline and the next progression during follow-up testing. Tests were administered online in class, with students given 40 min for each test which was overseen by members of the research team who were blinded to group allocation. Except for technological issues, no assistance was given to the students during the tests. Students who asked for assistance with a question were advised to skip to the next question and to return to it if they had time. To guard against contamination – to hold as much as possible constant – testing was conducted according to a strict timetable whereby all maths and reading tests were completed in the morning (across two consecutive days) and all science tests and questionnaires were completed in the afternoon.

Test scaled scores (interval scale from 0 to 120) were used for analysis. As ceiling effects limit the ability to draw inference about a student's improvement, students who answered all questions correctly at baseline were removed from the analysis for that subject (the difference between baseline and 8-month follow-up results).

2.3.2. Secondary outcomes

A range of secondary outcomes was also tested to gauge whether improvements in the quality of teaching as a result of participation in QTR also affected a broader range of social and emotional outcomes of schooling (Ladwig, 2010). Students completed a questionnaire to assess any changes in how they perceived their school life and educational and occupational aspirations.

Quality of School Life (QSL) was assessed using ACER's scales modified by Ainley and Bourke (1992), based on the original scales developed by Williams and Batten (1981). The QSL scale has a stable latent structure and the subscales have reasonable reliabilities (Mok & Mcdonald, 1994). Scale means are calculated for analysis, with scales analysed individually. These scales use a four-point measure (Agree, Mostly agree, Mostly disagree, Disagree) to gauge student perceptions of:

- General satisfaction (6 items): enjoyment in the school environment ($\alpha = 0.83$);
- Achievement (5 items): success as a student and ability to undertake class work ($\alpha = 0.83$);
- Teachers (5 items): teacher fairness and willingness to help ($\alpha = 0.86$);
- Relevance (6 items): schooling as useful for future endeavours ($\alpha = 0.85$); and
- Adventure (5 items): excitement and interest in class work ($\alpha = 0.84$).

Learning culture was assessed using a scale from the Aspirations Longitudinal Study Student Survey (Gore, Holmes, et al., 2015).

- Learning culture (5 items): level of support for and perceptions of the importance of study (four-point scale: Never, Rarely, Sometimes, Frequently; mean of items calculated for analysis).

2.3.3. Demographic data

Co-variates considered in the statistical models were: age, year level, gender, language background,[1] and Aboriginal and/or Torres Strait Islander status.[2]

2.3.4. Intervention fidelity

The nine features we considered to be essential to QTR and Peer Observation fidelity are outlined in Table 2. All PLCs were asked to provide regular data on implementation fidelity to evaluate whether interventions were being undertaken as designed and any adaptations made within and across settings. Intervention quality was monitored through: 1) fidelity check observations conducted by members of the research team who attended the entire session, and 2) fidelity reporting by at least one member of each PLC through an online survey for each round/peer observation session. The sum of fidelity items coded as complete for an intervention session (e.g., 5/9) as well as whether a session fulfilled all fidelity criteria (yes = 1, no = 0) were calculated for analysis.

2.3.5. Instructional volume

The average time per week dedicated to each subject area was

[1] Identifying students who speak a language other than English at home.
[2] Students who are First Nations peoples of Australia.

Table 2
Intervention fidelity checklists.

QTR and QTR-T	Peer Observation
1. Was a professional reading session conducted?	1. Did PLC members have a pre-observation meeting?
2. Was a full lesson observed?	2. Did you identify the focus of your observation?
3. Were all PLC members in attendance throughout the lesson?	3. Did you share the background and context for the observed lesson?
4. Did all PLC members individually code prior to discussion for this Round?	4. Were PLC members present throughout the lesson?
5. Did all PLC members provide their codes and justification (using lesson evidence) for each QT element?	5. How long after the observation did the lesson debrief occur? (<120 min required for fidelity)
6. Did PLC members take turns leading the discussion of elements during this Round?	6. Were all members of the PLC present for the post-observation debrief?
7. Was the QT Classroom Practice Guide a consistent point of reference throughout the discussion?	7. How long did the post-observation debrief last? (>30 min required for fidelity)
8. Were PLC members (including the observed teacher) present throughout the discussion?	8. Did the observer pose questions to prompt further development?
9. How long was the post lesson discussion? (>60 min required for fidelity)	9. Were the next steps planned?

investigated using a teacher questionnaire. Completed at the 8-month follow-up point, teachers were asked "*How many hours a week on average do your students spend learning the following subjects (to the nearest hour): for numeracy (mathematics), literacy (reading), reading for comprehension, and science?*" Reading for comprehension was included as a subset of literacy because the reading test largely focuses on this capability.

2.4. Sample size

Sampling calculations were based on comparison of the primary outcomes in students (i.e., PAT scores) for the Researcher-led QTR intervention and the PD-as-usual control arm and extended to the additional arms of the study (Trainer-led QTR and Peer Observation) based on this initial calculation. In order to detect a significant effect ($d = 0.20$) of around 3 months' additional growth (Higgins et al., 2016), approximately 788 students were required with 80% power and alpha set at 0.05. To adjust for the hierarchical structure of the data, the following correction factor was applied [1+ (m − 1) x ICC] (Donner & Klar, 2000), where m = students per class and ICC = the intra-class correlation coefficient. Assumptions were based on the hierarchical levels of students within classes and classes within schools (two classes per school, with approximately 20 students per class). A conservative ICC of 0.38 was assumed, based on the combined partitioning of variance at the between-school and between-class levels (Lamb & Fullarton, 2001), resulting in a correction factor of 8.22. The resulting required student sample was 6477 students at 161 schools for a minimum detectable effect size of $d = 0.20$. Extended across the four arms of the study the total sample was 12,954 students from 644 teachers' classes at 322 schools.

2.5. Randomisation

The school was the unit of randomisation in this study. Consenting schools were stratified based on school location (i.e., urban or rural) and socio-economic status using the continuous variable Index of Community Socio-Educational Advantage (ICSEA[3]). These two stratification variables were selected due to their use in classifying schools nationally and their availability on the Australian Government *My School* website (https://www.myschool.edu.au/). For school location, schools classified as Remote, Very Remote, Outer Regional, and Inner Regional were grouped into the 'rural' strata, while schools classified as Major Cities were grouped into the 'urban' strata. For socio-economic status, schools within each school location strata were ranked by their ICSEA and grouped into blocks of four (e.g., ICSEA 1, 2, 3, 4 = block 1, ICSEA 5, 6, 7, 8 = block 2).

Following stratification and blocking, schools within blocks were allocated to one of four conditions by an external researcher using a computerised random number generator. Schools within allocation blocks were ranked ascendingly by their random number, with allocation of groups by rank (1 = Researcher-led QTR; 2 = Trainer-led QTR; 3 = Peer Observation; and 4 = Wait-list control). Where blocks had fewer than four schools, the external researcher included dummy schools to complete the block and maintain the same probability of schools being allocated to one of the four conditions. Schools were notified of their group allocation as soon as possible after completion of baseline data collection at the school. This accords with the preferred method of randomisation in cluster randomised trials (Murray et al., 2004).

2.6. Data analysis

Statistical analyses of the primary and secondary outcomes were conducted using IBM PASW Statistics 25 (SPSS Inc. Chicago, IL) software. Impacts were estimated using an intention-to-treat (ITT) approach, with alpha levels set at $p < 0.05$. Linear mixed models were fitted to compare continuous outcomes for each of the intervention groups (Researcher-led QTR, Trainer-led QTR, and Peer Observation) against those of the PD-as-usual control condition. Group, time, and group-by-time interactions were assessed as fixed effects within the models. Co-variates of gender and year level (Year 3 or Year 4) were not found to be significant ($p > 0.05$) as fixed effects in the initial model and were removed in favour of testing the most parsimonious model. A repeated measures statement was included to model the within-subject correlated errors across time, and random intercepts were included for school and class nested within school to account for the hierarchical nature of the data. Differences in means and 95% confidence intervals (CIs) were determined using the linear mixed models, and the PD-as-usual control group was set as the comparison group for group-by-time contrasts. No additional treatment group contrasts were planned *a priori*, and no adjustments for multiplicity were applied to the group-by-time contrasts.

Hedges' g was used to determine effect sizes of the change in mean score for each group relative to the baseline value (effect of intervention on the mean change score), where $\bar{x}_1$ is the conditional estimate of the control group, and $\bar{x}_2$ is the conditional estimate of the intervention group being compared.

[3] ICSEA refers to the Index of Socio-Educational Advantage used in Australian school systems. The index includes socioeconomic measures, the geographical location of school and percentage of students from Indigenous backgrounds.

$$g = \frac{\bar{x}_2 - \bar{x}_1}{s^*}$$

s^* is estimated from the analysis sample as follows:

$$s^* = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

where $n1$ is the sample size in the control group, $n2$ is the sample size in the treatment group, $s1$ is the standard deviation of the control group, and $s2$ is the standard deviation of the treatment group (all estimates of standard deviation used are unconditional).

Ninety-five per cent confidence intervals (95% CIs) of the effect size were computed using the compute.es function (AC Del Re, 2013) in R version 3.4.4 (R Core Team, 2019). This function computes the confidence intervals using the variance in g derived by the Hedges and Olkin (1985, p. 86) formula:

$$var(g) = \frac{n_1 + n_2}{n_1 n_2} + \frac{g^2}{2(n_1 + n_2)}$$

2.7. Independent oversight

To ensure that our evaluation processes aligned with CONSORT guidelines, a senior analyst from the RAND Corporation was employed to provide independent oversight of the trial. This role included:

1. Monitoring the randomisation processes, including all correspondence between researchers and the statistician (external researcher) during the randomisation process and checking that alignment of allocation was maintained between baseline and follow-up datasets;
2. Ongoing advice to the project team to ensure data collection was compliant with CONSORT guidelines, including blinding and allocation concealment (i.e., allocation reveal post baseline data collection);
3. Ongoing advice to the project team to ensure minimisation of bias throughout the study, including split cohort design and decision-making following interim analysis; and
4. Independent review of the statistical procedures in line with statistical protocols for the primary outcomes, including access to original (de-identified) primary outcome datasets from ACER; alignment of 8-month follow-up and baseline datasets; independent review of the appropriateness of statistical modelling; and, confirmation of primary outcome results.

3. Results

The flow of participants through the study is reported in Fig. 1. The sample for the first cohort — the focus of this paper — consisted of 134 eligible schools making up 126 PLCs. Included were four small school networks (2 x 4 schools, 1 x 3 schools, 1 x 2 schools). One school withdrew from the study post-randomisation but prior to baseline data collection. Our final sample included 125 PLCs involving 5478 students from 234 classes.

Baseline data collection and randomisation occurred during Term 1 (February—March), with follow-up data collection during Term 4 (November—December) in 2019. Teachers at schools allocated to the intervention conditions (QTR, QTR-T, and Peer Observation) attended their respective two-day workshops in the first two weeks of Term 2 (May), with the in-school component of the interventions completed prior to the end of Term 2 (early July) by all intervention groups, with the exception of three schools in the QTR group, which completed early in Term 3 (late July).

Sample characteristics are displayed by allocation group in Table 3. At the school level, randomisation produced a balanced sample in terms of socio-educational advantage (ICSEA) (ACARA, 2015) and location of schools (rural/urban). Monitored teachers held an average of 10 years' (SD = 8.7) teaching experience, with marginal differences across groups for experience and qualifications. The control group included a slightly lower (~5%) proportion of Indigenous students compared to the other groups, and the proportion of students with a language background other than English (LBOTE) was lowest in the QTR group.

During the study, overall student attrition was low using the What Works Clearinghouse Standards (2020). When considering attrition relative to the control group, total attrition (combined control and selected other group) ranged from 15.0% to 21.6% across mathematics, reading and science outcomes, with differential attrition ranging from 0.3% to 3.4%. Attrition was highest for the QTR group, primarily because bushfires prevented follow-up data collection in one school.

3.1. Intervention fidelity

The mean scores for observed and self-reported fidelity, and the proportion of QTR or Peer Observation PD sessions that coded 100% fidelity (9/9) are detailed in Table 4. Fidelity was highest among the QTR group, followed by the QTR-T and Peer Observation groups respectively, which both reported substantially lower fulfilment of all criteria. Observed fidelity was marginally lower than self-reported fidelity for the QTR-T and Peer Observation groups.

3.2. Instructional volume

Teachers reported providing the largest volume of instruction in reading, followed by mathematics (Table 5). Reported time spent in reading for comprehension, as a specific reading focus, was approximately half that of the reported time spent in mathematics instruction, across all groups. Science was reported as receiving the least instructional time across all groups. Intervention groups (compared to the control group) displayed marginally greater time spent in all curriculum areas other than science.

3.3. Primary outcomes

Details of the primary outcomes are displayed in Table 6. Baseline values for mathematics, reading, and science were compared (using the fixed effect of group) and found to be equivalent across all groups ($p > 0.05$). For mathematics, a significant group-by-time effect ($p < 0.000$) was observed, with group-level contrasts (control as reference group) revealing a significant beneficial treatment effect for the QTR group ($p < 0.000$). Growth in mathematics achievement for the QTR group was 1.55 points higher on average than the control group across the intervention period. This equated to an effect size of 0.12 (95% CI: 0.07—0.17), which is equivalent to two months of additional growth (Education Endowment Foundation, 2018) across the 8-month intervention period. The QTR-T and Peer Observation intervention groups did not display significant treatment effects for mathematics.

There was a significant group-by-time interaction effect displayed for the reading outcome ($p = 0.004$) whereby group level contrasts revealed significantly less growth in the reading outcome among the Peer Observation group when compared to the Control group ($g = -0.07$; 95% CI = -0.13, -0.02; $p = 0.009$). The greatest growth among the three intervention groups was observed for the

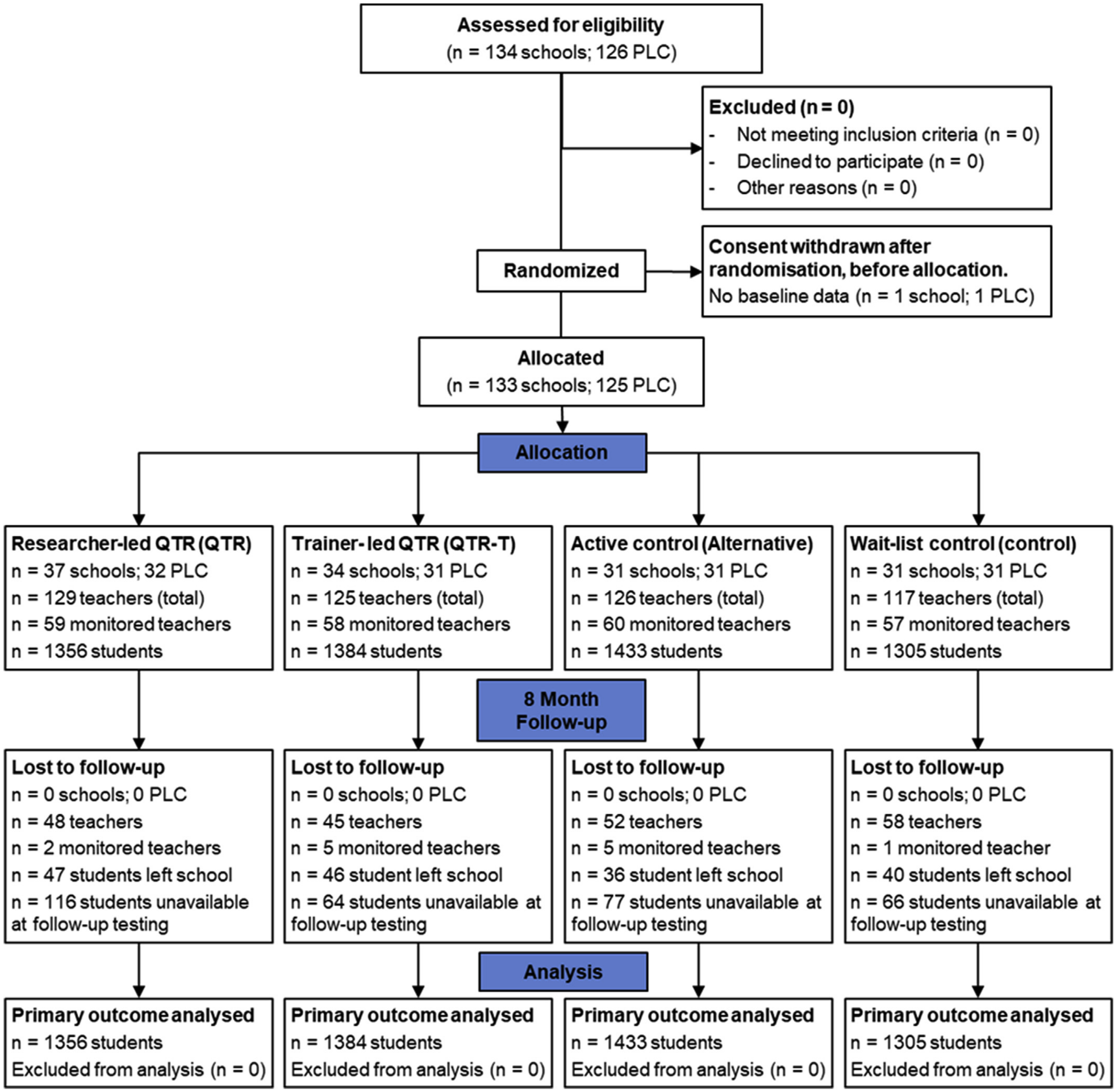

Fig. 1. Consort diagram.

QTR condition; however, the effect relative to the control group was not significant ($g = 0.02$; 95% CI = −0.03, 0.08; $p = 0.398$). There was no significant group-by-time interaction displayed for the science outcome ($p = 0.645$).

3.4. Secondary outcomes

Of the secondary outcomes (Table 7), only the achievement scale in the Quality of School Life questionnaire displayed a significant group-by-time interaction ($p = 0.003$) whereby a significant beneficial treatment effect was found for students in the Peer Observation group ($g = 0.14$; 95% CI = 0.04, 0.24; $p = 0.007$).

4. Discussion

Despite intensive research efforts over many years, the field still has limited understanding of what forms of teacher PD lead to changes in teachers' practice and, subsequently, to improvements in student achievement (Kennedy, 2016). The aim of this study was to evaluate the efficacy of the QTR PD intervention for the improvement of student outcomes, and to establish efficacy in relation to a scalable version of QTR (QTR-T) and a more common alternative (Peer Observation) PD intervention. By comparing multiple forms of PD intervention with PD-as-usual, the study has yielded several fresh insights.

Most importantly, our study demonstrates that QTR has

Table 3
Sample characteristics (Australia from March to November 2019).

Characteristics	QTR	QTR-T	Peer Observation	Control	Overall
Schools, *n*	32	31	31	31	125
ICSEA, mean (SD)	997.1 (84.7)	991.2 (79.3)	991.5 (89.1)	997.7 (77.1)	994.4 (81.7)
Rural, *n* (%)	13 (40.6)	14 (45.2)	13 (41.9)	14 (45.2)	54 (43.2)
Monitored teachers, *n*	**59**	**58**	**60**	**57**	**234**
Experience – years, mean (SD)[a]	10.6 (8.5)	12.4 (8.5)	9.2 (8.0)	11.2 (9.6)	10.8 (8.7)
Qualifications – Bachelor, %[b]	78.2	76.8	75.9	78.7	77.3
Qualifications – Masters, %[b]	16.4	12.5	19.0	14.9	15.7
Students, *n*	**1356**	**1384**	**1433**	**1305**	**5478**
Age – years, mean (SD)	9.7 (0.6)	9.7 (0.7)	9.8 (0.6)	9.7 (0.7)	9.7 (0.7)
Female, *n* (%)	690 (50.9)	680 (49.1)	707 (49.3)	647 (49.6)	2724 (49.7)
Indigenous, *n* (%)	129 (9.5)	113 (8.2)	120 (8.4)	65 (5.0)	427 (7.8)
LBOTE[c], *n* (%)	148 (10.9)	243 (17.6)	333 (23.2)	333 (25.5)	1057 (19.3)

[a] Based on valid survey responses: QTR, $n = 59$; QTR-T, $n = 56$; Peer Observation, $n = 57$; Control, $n = 48$.
[b] Based on valid survey responses: QTR, $n = 55$; QTR-T, $n = 56$; Peer Observation, $n = 56$; Control, $n = 46$.
[c] LBOTE = Language background other than English.

Table 4
Fidelity outcomes by group (Australia from March to November 2019).

Outcome	QTR	QTR-T	Peer Observation
Fidelity score			
Observed, mean (SD)	8.7 (0.9)	8.1 (1.5)	7.4 (1.5)
Self-reported, mean (SD)	8.8 (0.5)	8.5 (1.2)	8.1 (1.5)
Fidelity 9/9			
Observed, %	81.3	54.8	35.7
Self-reported, %	68.8	64.5	37.0

significant impact on students' mathematics achievement. Students of teachers in the QTR group displayed an additional two months' growth ($g = 0.12$) in mathematics relative to the control group across the eight-month intervention period. This finding is especially notable given the limited external input and relatively short duration of the intervention, both of which have positive implications for scalability. The magnitude of the effect is also impressive given a recent review of research on PD designed to improve student outcomes in elementary mathematics which found, across 12 rigorous studies, an average effect size of 0.04, with none reporting significant outcomes (Pellegrini et al., 2018). Other studies, primarily centred on content-focused PD, have also reported small or inconsistent effects of PD on student achievement in mathematics (see Jacob et al., 2017; Prast et al., 2018; Ross et al., 2006), demonstrating the ongoing challenge for the field in designing powerful PD programs that significantly increase student outcomes. The strong result for QTR's effect on student achievement in mathematics is a major contribution in identifying PD with demonstrable impact on student outcomes.

While the finding that student outcomes in reading and science were not significantly greater for the QTR intervention groups is somewhat disappointing in this light, it might reasonably be explained by instructional volume. Average instructional volume was 6.7 h per week for mathematics compared with 3.3 h per week for 'reading for comprehension' (the focus of the PAT-R assessment) and only 1.7 h per week for 'science.' We argue that if student achievement is related to the quality of teaching, which is enhanced by participation in QTR, the growth in student achievement in mathematics might be attributable to the greater level of exposure to quality teaching in mathematics.

Students in the QTR group received less than half of the instructional time in reading comprehension with a commensurate level of growth–approximately half of that achieved in mathematics. While other studies have found evidence that PD can have a significant effect on student achievement in reading (see Amendum, 2014; Garet et al., 2008; Porche et al., 2012), they primarily refer to content-focused PD that targets specific reading strategies (such as word attack, phonemes, or fluency). In addition, they use targeted testing of individual reading concepts, rather than standardised testing, to identify growth. Moreover, unlike QTR, reading interventions with the largest effects require substantial ongoing support by external coaches to achieve significant gains in student achievement (Amendum, 2014; Fine & Kossack, 2002).

Despite the strong positive finding for the QTR intervention, the effectiveness of the QTR-T intervention, the more scalable form of PD led by Advisers, was less clear. Taking high quality effective PD to scale is a key challenge for education systems (Coburn, 2003) and notoriously difficult primarily due to contextual variation and the large and dynamic workforce in education (Clarke & Dede, 2009; Kraft et al., 2018). When PD is taken to scale, it often fails to produce outcomes in student achievement (Kraft et al., 2018).

Our findings suggest that QTR Adviser experience and confidence in workshop delivery appears to have had a direct effect on the fidelity with which QTR was undertaken by participants in the QTR-T group and, in turn, on student outcomes. Given that our QTR Advisers have since had additional time to build their experience and confidence in QTR workshop delivery, the inclusion of a second cohort in this four-arm trial (results available early in 2022) provides an ideal opportunity to advance our knowledge of an effective model of QTR delivery that can be taken to scale.

Current literature within the fields of PD and school improvement clearly point to the potential effectiveness of classroom

Table 5
Instructional volume. (Australia from March to November 2019).

	QTR	QTR-T	Alternative	Control
Mathematics hrs/wk, mean (SD) [a]	7.5 (2.4)	6.3 (2.3)	7.1 (3.2)	6.2 (3.8)
Reading hrs/wk, mean (SD) [a]	9.7 (1.7)	9.0 (2.5)	9.2 (3.8)	7.9 (3.6)
Reading comprehension hrs/wk, mean (SD) [a]	3.5 (1.8)	3.1 (1.2)	3.8 (2.1)	2.8 (2.1)
Science hrs/wk, mean (SD) [a]	1.6 (0.7)	1.3 (0.5)	2.1 (2.0)	1.9 (1.1)

[a] Based on valid survey responses: QTR, $n = 46$; QTR-T, $n = 50$; Alternative, $n = 42$; Control, $n = 35$.

Table 6
Primary outcomes comparison by group - Baseline to 8 months; Intention-to-treat analysis (Australia from March to November 2019).

Outcome	n	Baseline, mean (95% CI)	Ceiling, n (%)	Retest %	n (miss)	Mean change from baseline (95% CI)	Adjusted mean difference (95% CI)[a]	Adjusted effect size g (95% CI)[a]	Group by time p	Post hoc p
Mathematics										
QTR	1307	109.07 (107.13, 111.01)	6 (0.4)	89	1165 (142)	7.76 (7.28, 8.24)	1.55 (0.87, 2.23)	0.12 (0.07, 0.17)	0.000*	0.000*
QTR-T	1329	109.30 (107.35, 111.25)	11 (0.8)	93	1230 (99)	6.43 (5.96, 6.89)	0.22 (−0.45, 0.89)	0.02 (−0.03, 0.07)		0.527
Peer Observation	1387	109.92 (107.98, 111.85)	6 (0.4)	91	1269 (118)	6.81 (6.35, 7.27)	0.60 (−0.06, 1.27)	0.05 (0.00, 0.10)		0.075
Control	1258	109.80 (107.84, 111.77)	11 (0.8)	92	1161 (97)	6.21 (5.73, 6.69)	Reference	Reference		
Reading										
QTR	1323	106.11 (103.97, 108.24)	2 (0.2)	89	1171 (152)	9.01 (8.4, 9.62)	0.37 (−0.49, 1.23)	0.02 (−0.03, 0.08)	0.004*	0.398
QTR-T	1332	105.65 (103.5, 107.81)	5 (0.4)	93	1240 (92)	8.39 (7.79, 8.98)	−0.25 (−1.11, 0.6)	−0.02 (−0.07, 0.04)		0.561
Peer Observation	1402	106.83 (104.7, 108.96)	5 (0.4)	91	1274 (128)	7.51 (6.93, 8.1)	−1.12 (−1.97, −0.28)	−0.07 (−0.13, −0.02)		0.009*
Control	1272	105.33 (103.16, 107.5)	2 (0.2)	92	1169 (103)	8.64 (8.03, 9.25)	Reference	Reference		
Science										
QTR	1297	109.08 (107.69, 110.47)	7 (0.5)	88	1136 (161)	3.75 (3.26, 4.23)	−0.13 (−0.81, 0.56)	−0.01 (−0.08, 0.05)	0.645	0.714
QTR-T	1268	108.64 (107.24, 110.04)	7 (0.5)	93	1181 (87)	3.75 (3.26, 4.24)	−0.13 (−0.81, 0.56)	−0.01 (−0.08, 0.05)		0.719
Peer Observation	1353	109.15 (107.76, 110.54)	5 (0.3)	93	1253 (100)	3.45 (2.98, 3.92)	−0.42 (−1.1, 0.25)	−0.04 (−0.11, 0.02)		0.217
Control	1265	108.67 (107.26, 110.08)	6 (0.5)	91	1149 (116)	3.87 (3.39, 4.36)	Reference	Reference		

* Significance at $p < 0.05$.
[a] Between-group difference of change score (intervention change minus control change).

observation and professional learning communities for promoting ongoing change in teacher practice (Giles & Hargreaves, 2006; Goddard, Skrla & Salloum, 2017; Jones & Thessin, 2017). Our study provides clear evidence that teachers' engagement in collaborative PLCs that undertake observations of each other's teaching is not sufficient to ensure growth in student achievement. Teachers who were allocated to the Peer Observation intervention arm engaged in precisely the types of collaborative processes of observing, reflecting, receiving feedback on, and improving classroom practice that are frequently advocated by scholars in the PD field (Glen et al., 2017; Timperley 2011). Furthermore, these teachers were provided with the same amount of time and same funding for release as those in the QTR and QTR-T arms. However, the students in these schools did not achieve the same level of growth as their counterparts in the QTR and QTR-T groups. This result provides empirical evidence in support of Kennedy's (2016) observation that:

> As researchers, we need to move past the concept of learning communities per se and begin examining the content such groups discuss and the nature of the intellectual work they are engaged in. (p. 972)

QTR's focus on substance and not just process and its mechanisms for engaging teachers in intellectual analysis of pedagogy arguably sets it apart from many other forms of collaborative PD (Bowe & Gore, 2017). Teachers in the QTR and QTR-T groups framed their observations and discussions with the rigorously tested Quality Teaching model (NSW DET, 2006), which provided them with a comprehensive vision of good teaching (Gore, in press) and shared language for analysing and discussing the pedagogy they observed.

4.1. Limitations

Despite the rigour of this four-arm RCT, some limitations should be noted. First, these significant results were obtained from approximately half of the desired sample. Data from the second (post-COVID-19) cohort, planned for Terms 1 and 4 of 2021, will provide an opportunity to reassess findings with a larger sample. Second, while the sample as randomised was balanced across the study arms, and broadly representative of Australian government schools, it may not be representative of schools within the Catholic and independent sectors. Finally, while participants were blind to treatment allocation until completion of baseline data collection, they were not blind to allocation during the intervention period. The extent to which business-as-usual was maintained for mathematics, reading, and science instruction among control group schools was not assessed. Furthermore, while we were not aware of concurrent programs focused on mathematics, reading, and science outcomes, we cannot disentangle any impact from school-based agendas across the intervention period.

5. Conclusion

Significant results from RCTs in the field of PD are rare. This study has demonstrated that QTR has a positive effect on student outcomes in mathematics and is showing promising results for outcomes in reading. In combination with results of earlier studies showing positive effects on the quality of teaching and teacher morale, QTR is, demonstrably, a form of PD with considerable potential for making the kind of difference sought by policy makers, governments, educators, and the community. Our study is to be repeated with a second cohort of teachers and students which will enable further testing of the results reported here. However, postponement of that work due to the COVID-19 pandemic will

Table 7
Changes in secondary outcomes (Student) from baseline to 8 months (Australia from March to November 2019).

Outcome	Group	Baseline		8-month		Within group[a] p	Group x time p	Post hoc: Time; Adjusted effect size g (95% CI)[a]; p
		n	mean (95% CI)	n	mean (95% CI)			
General satisfaction	QTR	1082	1.64 (1.58, 1.70)	833	1.76 (1.69, 1.82)	0.000	0.162	NSI
	QTR-T	1079	1.60 (1.54, 1.66)	874	1.78 (1.72, 1.85)	0.000		NSI
	Peer Observation	1126	1.56 (1.50, 1.62)	906	1.72 (1.65, 1.78)	0.000		NSI
	Control	1054	1.63 (1.57, 1.69)	832	1.75 (1.69, 1.82)	0.000		reference
Achievement	QTR	1082	1.56 (1.52, 1.61)	833	1.57 (1.53, 1.62)	0.660	0.003	NSI
	QTR-T	1078	1.54 (1.50, 1.58)	874	1.60 (1.56, 1.65)	0.001		NSI
	Peer Observation	1114	1.54 (1.49, 1.58)	906	1.64 (1.60, 1.69)	0.000		B-8M; 0.14 (0.04, 0.24); 0.007
	Control	1051	1.57 (1.53, 1.61)	832	1.60 (1.56, 1.65)	0.106		reference
Teachers	QTR	1082	1.40 (1.35, 1.45)	833	1.43 (1.38, 1.48)	0.223	0.148	NSI
	QTR-T	1077	1.36 (1.31, 1.41)	874	1.44 (1.38, 1.49)	0.000		NSI
	Peer Observation	1117	1.35 (1.30, 1.40)	906	1.45 (1.39, 1.50)	0.000		NSI
	Control	1050	1.40 (1.35, 1.45)	832	1.45 (1.40, 1.51)	0.011		reference
Relevance	QTR	1078	1.36 (1.32, 1.40)	833	1.34 (1.30, 1.38)	0.263	0.195	NSI
	QTR-T	1078	1.35 (1.31, 1.39)	874	1.36 (1.32, 1.41)	0.388		NSI
	Peer Observation	1111	1.34 (1.30, 1.38)	906	1.37 (1.33, 1.42)	0.056		NSI
	Control	1052	1.36 (1.32, 1.40)	832	1.37 (1.33, 1.41)	0.499		reference
Adventure	QTR	1084	1.83 (1.77, 1.90)	833	1.96 (1.89, 2.03)	0.000	0.163	NSI
	QTR-T	1075	1.79 (1.72, 1.85)	874	1.99 (1.92, 2.05)	0.000		NSI
	Peer Observation	1112	1.78 (1.71, 1.84)	906	1.97 (1.90, 2.04)	0.000		NSI
	Control	1046	1.82 (1.75, 1.89)	832	1.99 (1.92, 2.06)	0.000		reference
Learning culture	QTR	1073	2.75 (2.71, 2.78)	830	2.76 (2.72, 2.79)	0.681	0.933	NSI
	QTR-T	1071	2.77 (2.74, 2.81)	872	2.79 (2.75, 2.82)	0.404		NSI
	Peer Observation	1100	2.75 (2.72, 2.78)	903	2.75 (2.71, 2.78)	0.927		NSI
	Control	1043	2.74 (2.71, 2.78)	830	2.75 (2.71, 2.78)	0.733		reference

Note. * Significance at $p < 0.05$. NSI = No significant interaction
[a] Between-group difference of change score (intervention change minus control change).

mean additional results will not be available until early in 2022. While at this stage we only have results for a single cohort, these very promising results are too important for the field to wait nearly two years before publishing.

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Appendix A. Primary outcomes comparison by group (means and standard deviations) - Baseline to 8 months; Intention-to-treat analysis (Australia from March to November 2019)

Outcome	n	Baseline mean (SD)	Ceiling, n (%)	Retest %	n (miss)	8-months mean (SD)	Group x time p	
Mathematics								
QTR	1307	109.52 (13.04)	6 (0.4)	89	1165 (142)	117.39 (12.48)	0.000*	0.000*
QTR-T	1329	109.35 (13.41)	11 (0.8)	93	1230 (99)	115.97 (12.20)		0.527
Peer Observation	1387	110.31 (13.44)	6 (0.4)	91	1269 (118)	117.27 (12.70)		0.075
Control	1258	110.24 (12.73)	11 (0.8)	92	1161 (97)	116.58 (11.88)		Reference
Reading								
QTR	1323	106.71 (15.60)	2 (0.2)	89	1171 (152)	115.93 (14.20)	0.004*	0.398
QTR-T	1332	105.74 (15.82)	5 (0.4)	93	1240 (92)	114.17 (14.00)		0.561
Peer Observation	1402	107.32 (15.53)	5 (0.4)	91	1274 (128)	114.96 (13.37)		0.009*
Control	1272	105.89 (15.29)	2 (0.2)	92	1169 (103)	114.64 (13.41)		Reference
Science								
QTR	1297	109.33 (10.9)	7 (0.5)	88	1136 (161)	113.2 (9.43)	0.645	0.714
QTR-T	1268	108.85 (11.36)	7 (0.5)	93	1181 (87)	112.27 (9.49)		0.719
Peer Observation	1353	109.36 (10.47)	5 (0.3)	93	1253 (100)	112.99 (9.34)		0.217
Control	1265	108.9 (10.67)	6 (0.5)	91	1149 (116)	112.9 (9.13)		Reference

* Significance at $p < 0.05$.
[a] Between-group difference of change score (intervention change minus control change).

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